

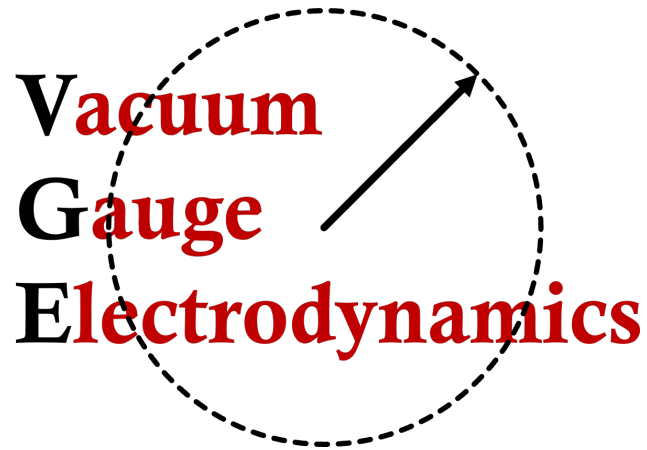
Weyl Neutrinos
and
Electron Coulomb Fields

Dr. Christopher Bradshaw Hayes

May 14, 2026

Abstract

In the spherically based coordinate system, massless Weyl neutrinos assume the role of radiated (or emitted) particles. As all known neutrinos are produced in particle decays, this provides a natural platform for the discussion of Weyl neutrinos. Unfortunately, it also presents a rather significant complication for the theory of massive Dirac particles, which become sources of Weyl neutrino radiation. Essentially, Weyl neutrinos become linked to particle Coulomb fields.



Contents

1	Weyl Neutrinos	4
1.1	Weyl Neutrino's in the Spherical Basis	4
1.2	Two-Component Problem for Weyl Neutrinos	5
2	Electron Coulomb Fields and Weyl Neutrinos	8
2.1	Construction of Electromagnetic Eigenstates	8
2.2	Inverse Transformation and Unitary Matrices	10
3	Hamiltonians for Weyl Neutrinos	12
4	Electromagnetic Noether Currents	14

List of Figures

1	Spacetime diagrams for Weyl neutrinos in the spherical basis	6
---	--	---

List of Tables

1	Solutions to the Weyl equation in the spherical basis	4
2	Unitary matrices for Weyl neutrinos	8

1 Weyl Neutrinos

When the mass term in the free particle Dirac equation is set to zero, the resulting eigenvalue problem simplifies to

$$i\hbar c[\gamma^\nu \partial_\nu] \psi = 0 \quad (1.1)$$

Operating in a Cartesian coordinate system—and with chiral representation gamma matrices—this equation de-couples into a set of relativistic Weyl equations,

$$i\hbar c[\partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}] \psi_L = 0 \quad i\hbar c[\partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}] \psi_R = 0 \quad (1.2)$$

Two-component solutions to each equation are plane waves referred to as left- and right-handed Weyl spinors ψ_L and ψ_R . With energies $E = pc$, these spinors describe massless Weyl neutrinos. Weyl neutrinos are not their own anti-particles, implying particle and anti-particle spinors are derivable from each other through the charge conjugation operation.

1.1 Weyl Neutrino's in the Spherical Basis

The free particle Dirac equation in the spherical basis is

$$i\hbar c \left[\gamma^\nu \beta_\nu \partial_\tau - \gamma^\nu U_\nu \partial_\rho - \frac{1}{R} \gamma^\nu \vartheta_\nu \partial_\theta - \frac{1}{R \sin \theta} \gamma^\nu \varphi_\nu \partial_\phi \right] \psi - mc^2 \psi = 0 \quad (1.3)$$

Setting the mass term to zero—and assuming solutions do not depend on the coordinates (θ, ϕ) —determines

$$i\hbar c [\gamma^\nu \beta_\nu \partial_\tau - \gamma^\nu U_\nu \partial_\rho] \psi = 0 \quad (1.4)$$

This equation is the spherically based version of (1.1). Travelling wave solutions are available in Table 1 exhibiting complex exponentials written in terms of coordinates $(c\tau, \rho)$, and also in terms of retarded and advanced times (ct_r, ct_a) . Regardless of how

Particle	Column A	Column B
Neutrino	$\psi = \psi_o e^{-ik(c\tau - \rho)} = \psi_o e^{-ikct_r/\gamma}$	$\psi = \psi_o e^{-ik(c\tau + \rho)} = \psi_o e^{-ikct_a/\gamma}$
Anti-Neutrino	$\psi = \psi_o e^{ik(c\tau - \rho)} = \psi_o e^{ikct_r/\gamma}$	$\psi = \psi_o e^{ik(c\tau + \rho)} = \psi_o e^{ikct_a/\gamma}$

Table 1: *Solutions for Weyl neutrinos in the spherically-based coordinate system.*

exponentials are written, spinors ψ_o satisfy one of two momentum space equations,

$$\varepsilon [\gamma^\nu \beta_\nu + \gamma^\nu U_\nu] \psi_o = 0 \quad \text{or} \quad \varepsilon \left[\frac{\gamma^\nu R_\nu}{\rho} \right] \psi_o = 0 \quad (1.5a)$$

$$\varepsilon [-\gamma^\nu \beta_\nu + \gamma^\nu U_\nu] \psi_o = 0 \quad \text{or} \quad \varepsilon \left[\frac{\gamma^\nu R_\nu^a}{\rho_a} \right] \psi_o = 0 \quad (1.5b)$$

An understanding of these two formulas requires knowledge of the theory of retarded and advanced potentials¹. Defining $\boldsymbol{\varepsilon} \equiv \varepsilon \hat{\mathbf{n}} = \mathbf{p}c$, momentum space equations in terms of radiation basis gamma matrices, and Doppler four vectors are:

$$[\vec{\gamma} \cdot \vec{\varepsilon}_r] \psi_o = 0 \quad \vec{\varepsilon}_r = \frac{(\varepsilon, \boldsymbol{\varepsilon})}{\gamma(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta})} \equiv (\varepsilon_r, \boldsymbol{\varepsilon}_r) \quad (1.7a)$$

$$[\vec{\gamma} \cdot \vec{\varepsilon}_a] \psi_o = 0 \quad \vec{\varepsilon}_a = \frac{(\varepsilon, -\boldsymbol{\varepsilon})}{\gamma(1 + \hat{\mathbf{n}} \cdot \boldsymbol{\beta})} \equiv (\varepsilon_a, -\boldsymbol{\varepsilon}_a) \quad (1.7b)$$

From the second line, it is evident that the advanced time eigenvalue problem provides a description of Weyl neutrinos travelling in a direction opposite to the retarded time solution. For clarity, both Doppler four vectors are prepared graphically in the spacetime diagrams of Figure 1. The diagrams only indicate an inertial frame moving at velocity $\boldsymbol{\beta}$ and the emission of a Weyl neutrino moving along the forward light cone. There is no suggestion in either diagram of a parent particle that emits (or decays into) a Weyl neutrino.

1.2 Two-Component Problem for Weyl Neutrinos

Advanced time solutions to the Weyl equation are completely legitimate, but not required for discussions of the detailed properties of Weyl neutrinos. As such, a formulation of the two-component problem in the spherical basis can be addressed by focusing solely on the retarded problem. For this, use gamma-matrices in the chiral representation—but defined in terms of two-component Pauli 4-vectors:

$$\gamma^\nu = \begin{bmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{bmatrix} \quad (1.8)$$

¹Retarded and advanced potentials describe the propagation of electromagnetic fields using the retarded and advanced null vectors $R^\nu = (R, \mathbf{R})$ and $R_a^\nu = (R, -\mathbf{R})$. Essential properties of the null vectors can be summarized by writing

$$R^\nu = \rho[U^\nu + \beta^\nu] \quad \text{with} \quad \beta_\nu R^\nu = \rho \quad \text{and} \quad U_\nu R^\nu = -\rho \quad (1.6a)$$

$$R_a^\nu = \rho_a[\beta^\nu - U^\nu] \quad \text{with} \quad \beta_\nu R_a^\nu = \rho_a \quad \text{and} \quad U_\nu R_a^\nu = \rho_a \quad (1.6b)$$

See F. Rhorlich: “Classical Charged Particles” , 1965 for more information.

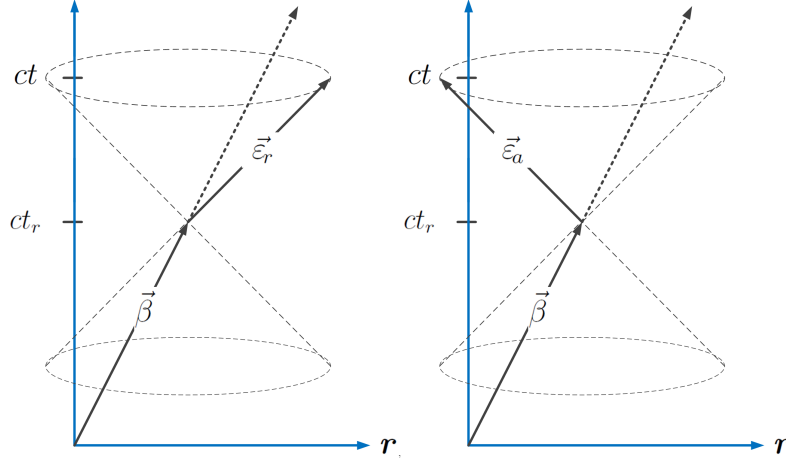


Figure 1: *Spacetime diagrams indicating Doppler solutions to the free particle Dirac equation for zero mass Weyl neutrinos in the spherical basis.*

Applying this operator generates de-coupled Weyl equations, with their time dependent solutions displayed on the right

$$i\hbar c [\sigma^\nu \beta_\nu \partial_\tau - \sigma^\nu U_\nu \partial_\rho] \psi_R = 0 \quad \psi_R(c\tau, \rho) = \psi_R \cdot e^{+ik(c\tau - \rho)} \quad (1.9a)$$

$$i\hbar c [\bar{\sigma}^\nu \beta_\nu \partial_\tau - \bar{\sigma}^\nu U_\nu \partial_\rho] \psi_L = 0 \quad \psi_L(c\tau, \rho) = \psi_L \cdot e^{-ik(c\tau - \rho)} \quad (1.9b)$$

The two first order equations are essentially the spherical basis counterpart of the Cartesian basis equations in (1.2), and lead to momentum space equations,

$$\varepsilon_r [1 - \boldsymbol{\sigma} \cdot \hat{\mathbf{n}}] \psi_R = 0 \quad \varepsilon_r (1 + \boldsymbol{\sigma} \cdot \hat{\mathbf{n}}) \psi_L = 0 \quad (1.10)$$

Similarities between (1.10) and the Cartesian-based momentum space equations are clearly evident, with the exception of a Doppler energy and a retarded unit vector $\hat{\mathbf{n}}$. A two-component Hamiltonian operator $\mathcal{H}_r$ is the same for each eigenvalue problem

$$\mathcal{H}_r = \varepsilon_r \boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \quad (1.11)$$

and features normalized eigenstates

$$\psi_R = N_1 \begin{bmatrix} 1 + n_z \\ n_x + in_y \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad \psi_L^c = N_1 \begin{bmatrix} -n_x + in_y \\ 1 + n_z \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad (1.12a)$$

$$\psi_L = N_2 \begin{bmatrix} 1 - n_z \\ -n_x - in_y \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad \psi_R^c = N_2 \begin{bmatrix} n_x - in_y \\ 1 - n_z \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (1.12b)$$

The $\pm$ -signs chosen for the exponentials indicates a consistency with the theory of massive Dirac neutrinos, which allows only left-handed neutrinos and right-handed

anti-neutrinos. Normalization factors ensuring $\psi^\dagger \psi = \varepsilon_r$ for each spinor are

$$N_1 = \sqrt{\frac{\varepsilon_r}{2(1+n_z)}} \quad N_2 = \sqrt{\frac{\varepsilon_r}{2(1-n_z)}} \quad (1.13)$$

The right-handed states have positive energy eigenvalues while the left-handed states have negative energy eigenvalues. Eigenstates with upper-case^c labels indicate charge conjugated states. These labels can be verified with the charge conjugation operator $\mathbb{C}$, implemented by the formula

$$\begin{bmatrix} \psi_L^c \\ \psi_R^c \end{bmatrix} = \eta_c C [\bar{\psi}]^T = \begin{bmatrix} -i\sigma^2 \psi_R^* \\ i\sigma^2 \psi_L^* \end{bmatrix} \quad \text{where} \quad C = - \begin{bmatrix} 0 & i\sigma_y \\ i\sigma_y & 0 \end{bmatrix} \quad (1.14)$$

and where C is the charge conjugation matrix. The application of $\mathbb{C}$ to each eigenstate determines

$$\mathbb{C}[\psi_R] = \psi_L^c \quad \mathbb{C}[\psi_L] = -\psi_R^c \quad \mathbb{C}[\psi_R^c] = -\psi_L \quad \mathbb{C}[\psi_L^c] = \psi_R \quad (1.15)$$

and verifies the labels in equation (1.12). With $\mathbf{j} = \hbar \boldsymbol{\sigma}/2$, rotations of eigenstates about the direction of motion are determined by

$$\begin{cases} \psi_R' = e^{-i\phi/2} \cdot \psi_R \\ \psi_L' = e^{+i\phi/2} \cdot \psi_L \end{cases} \quad \text{where} \quad \mathcal{R}(\hat{\mathbf{n}}; \phi) = e^{-i\mathbf{j} \cdot \hat{\mathbf{n}} \phi / \hbar} \quad (1.16)$$

Dirac currents can be derived from the construction $J^\nu = \psi^\dagger \sigma^\nu \psi$. For each of the four spinors above one finds

$$J_R^\nu = \psi_R^\dagger \sigma^\nu \psi_R = \varepsilon_r(1, \mathbf{n}) \quad (J_R^\nu)^c = \psi_R^{\dagger c} \sigma^\nu \psi_R^c = \varepsilon_r(1, \mathbf{n}) \quad (1.17a)$$

$$J_L^\nu = \psi_L^\dagger \bar{\sigma}^\nu \psi_L = \varepsilon_r(1, \mathbf{n}) \quad (J_L^\nu)^c = \psi_L^{\dagger c} \bar{\sigma}^\nu \psi_L^c = \varepsilon_r(1, \mathbf{n}) \quad (1.17b)$$

Unfortunately, conserved currents do not follow from these formulas. Instead, the continuity equations requires a source

$$\partial_\nu J^\nu = \varepsilon \partial_\nu \left[\frac{R^\nu}{\rho} \right] = \frac{2\varepsilon}{\rho} \quad (1.18)$$

This is a clear break from the conventional theory, likely resulting from the re-interpretation of Weyl neutrinos as radiated particles.

Unitary Matrices: With proper normalizations, neutrino state vectors of equation (1.12) can be combined into a pair of 2×2 special unitary matrices. Keeping particles with their anti-particles, matrices u_1 and u_2 are defined as:

$$u_1 = \frac{1}{\sqrt{2(1+n_z)}} \begin{bmatrix} 1+n_z & -n_x + in_y \\ n_x + in_y & 1+n_z \end{bmatrix} \quad u_2 = \frac{1}{\sqrt{2(1-n_z)}} \begin{bmatrix} 1-n_z & n_x - in_y \\ -n_x - in_y & 1-n_z \end{bmatrix}$$

Matrix	Characteristic Equation	Eigenvalue 1	Eigenvalue 2
u_1	$\lambda^2 - 2\lambda \cos \theta/2 + 1 = 0$	$\lambda_1 = e^{i\theta/2}$	$\lambda_1^* = e^{-i\theta/2}$
u_2	$\lambda^2 - 2\lambda \sin \theta/2 + 1 = 0$	$\lambda_2 = -ie^{i\theta/2}$	$\lambda_2^* = ie^{-i\theta/2}$

Table 2: *Characteristic equation and eigenvalues associated with matrices u_1 and u_2 .*

Both matrices will diagonalize the Hamiltonian operator—summarized by writing

$$u_1^\dagger \mathcal{H}_r u_1 = \begin{bmatrix} \varepsilon_r & 0 \\ 0 & -\varepsilon_r \end{bmatrix} \quad u_2^\dagger \mathcal{H}_r u_2 = \begin{bmatrix} -\varepsilon_r & 0 \\ 0 & \varepsilon_r \end{bmatrix} \quad (1.20)$$

It is important to solve the eigenvalue problem for both matrices in (1.19)—most easily accomplished by writing them in terms of angular coordinates (θ, ϕ) . Both matrices feature the same set of orthonormal eigenstates

$$|\lambda\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} i \\ e^{i\phi} \end{bmatrix} \quad |\lambda\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} -i \\ e^{i\phi} \end{bmatrix} \quad (1.21)$$

Associated distinct eigenvalues and characteristic equations for u_1 and u_2 are displayed in Table 2.

2 Electron Coulomb Fields and Weyl Neutrinos

An interesting possibility directly associated with the spherically-based coordinate system is the ability to re-construct the entire theory of Weyl neutrinos in a basis of electromagnetic eigenstates. This includes calculations of charge conjugated states, unitary matrices, and Noether currents.

2.1 Construction of Electromagnetic Eigenstates

Employing classical electron Coulomb fields $(\mathbf{E}, \mathbf{B})$, a four-component (complexified) electromagnetic paravector can be defined by

$$\mathcal{Z}^\nu \equiv (\eta, \mathbf{E} + i\mathbf{B}) = (\eta, \mathcal{Z}) \quad (2.1)$$

Treating it as an authentic four vector one finds $\mathcal{Z}^\nu \mathcal{Z}_\nu = 0$. Components of this paravector are important because they are visible in electromagnetic eigenstates derived from the Weyl neutrino eigenstates in equation (1.12). This can be demonstrated by forming linear combinations of Weyl neutrino eigenstates using selectively chosen

components of particle velocity fields. As an example—and with neutrino normalizations changed to $N_1 = N_2 = 1$ —right-handed eigenstates can be combined as

$$E_z \psi_R + (E_x + iE_y) \psi_R^c = \begin{bmatrix} \eta + \mathcal{Z}_3 \\ \mathcal{Z}_1 + i\mathcal{Z}_2 \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (2.2a)$$

$$(\eta + iB_z) \psi_R + (iB_x - B_y) \psi_R^c = \begin{bmatrix} \eta + \mathcal{Z}_3 \\ \mathcal{Z}_1 + i\mathcal{Z}_2 \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (2.2b)$$

Now add the two equations together producing a single waveform employing all components of particle Coulomb fields plus the dilatation. This result, along with three other possibilities, is:

$$\Psi_R = \frac{1}{2} [(\eta + \mathcal{Z}_3) \psi_R + (\mathcal{Z}_1 + i\mathcal{Z}_2) \psi_R^c] = \begin{bmatrix} \eta + \mathcal{Z}_3 \\ \mathcal{Z}_1 + i\mathcal{Z}_2 \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (2.3a)$$

$$\Psi_L = \frac{1}{2} [(\eta - \mathcal{Z}_3^*) \psi_L - (\mathcal{Z}_1^* + i\mathcal{Z}_2^*) \psi_L^c] = \begin{bmatrix} \eta - \mathcal{Z}_3^* \\ -\mathcal{Z}_1^* - i\mathcal{Z}_2^* \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad (2.3b)$$

$$\Psi_R^c = \frac{1}{2} [(\mathcal{Z}_1 - i\mathcal{Z}_2) \psi_R + (\eta - \mathcal{Z}_3) \psi_R^c] = \begin{bmatrix} \mathcal{Z}_1 - i\mathcal{Z}_2 \\ \eta - \mathcal{Z}_3 \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (2.3c)$$

$$\Psi_L^c = \frac{1}{2} [(-\mathcal{Z}_1^* + i\mathcal{Z}_2^*) \psi_L + (\eta + \mathcal{Z}_3^*) \psi_L^c] = \begin{bmatrix} -\mathcal{Z}_1^* + i\mathcal{Z}_2^* \\ \eta + \mathcal{Z}_3^* \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad (2.3d)$$

Equations (2.3) are impressive². Each electromagnetic state vector contains all the information necessary to build an electromagnetic field, and is composed of two purely left-handed or right-handed neutrino states. Moreover, since neutrino eigenstates of the same helicity have the same eigenvalue, this means each state vector in equation (2.3) is still an eigenstate of the Hamiltonian $\mathcal{H}_r$ in (1.11). Assume (Ψ_R, Ψ_L) are associated with an electron field while (Ψ_R^c, Ψ_L^c) are associated with a positron field. Each particle is therefore associated with one left-handed state and one right-handed state. The handedness of the state is easily recognized too, because left-handed states are always marked by starred components of $\mathcal{Z}$. This associates handedness with the phase $e^{\pm i\pi/2}$ existing between the particles electric and magnetic fields. As suggested by the particle labels, one may apply the charge conjugation operator to

²Unnormalized eigenstates in equation (2.3) can be derived using Pauli 4-vectors applied to the four-gradient and vacuum gauge potentials:

$$\bar{\sigma}^\nu \mathcal{Z}_\nu = \bar{\sigma}^\nu \partial_\nu \sigma^\nu A_\nu \quad \sigma^\nu \mathcal{Z}_\nu^* = \sigma^\nu \partial_\nu \bar{\sigma}^\nu A_\nu \quad (2.4)$$

unambiguously connect particle states with anti-particles states. A set of equations similar to (1.15) for state vectors in (2.3) is:

$$\mathbb{C}[\Psi_R] = \Psi_L^c \quad \mathbb{C}[\Psi_L] = -\Psi_R^c \quad \mathbb{C}[\Psi_R^c] = -\Psi_L \quad \mathbb{C}[\Psi_L^c] = \Psi_R \quad (2.5)$$

2.2 Inverse Transformation and Unitary Matrices

Equations (2.3) clearly shows electromagnetic eigenstates as a superposition of neutrino eigenstates, but one may also solve for the neutrino eigenstates in terms of the electromagnetic states.

$$\psi_R = \frac{1}{2\eta} [(1 + n_z)\Psi_R + (n_x + in_y)\Psi_R^c] \quad (2.6a)$$

$$\psi_R^c = \frac{1}{2\eta} [(n_x - in_y)\Psi_R + (1 - n_z)\Psi_R^c] \quad (2.6b)$$

$$\psi_L = \frac{1}{2\eta} [(1 - n_z)\Psi_L - (n_x + in_y)\Psi_L^c] \quad (2.6c)$$

$$\psi_L^c = \frac{1}{2\eta} [-(n_x - in_y)\Psi_L + (1 + n_z)\Psi_L^c] \quad (2.6d)$$

Once again, the neutrino eigenstates are constructed from only those electromagnetic states of the same handedness. Moreover, scalar fields multiplying each wave function on the right are simply components of the neutrino eigenstates.

Pairing of Eigenstates: Matrix equations linking neutrino and electromagnetic eigenstates can also be written as a pair of 2×2 transformations. For the right-handed states only,

$$\begin{bmatrix} \Psi_R \\ \Psi_R^c \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \eta + \mathcal{Z}_3 & \mathcal{Z}_1 + i\mathcal{Z}_2 \\ \mathcal{Z}_1 - i\mathcal{Z}_2 & \eta - \mathcal{Z}_3 \end{bmatrix} \cdot \begin{bmatrix} \psi_R \\ \psi_R^c \end{bmatrix} \quad (2.7a)$$

$$\begin{bmatrix} \psi_R \\ \psi_R^c \end{bmatrix} = \frac{1}{2\eta} \begin{bmatrix} 1 + n_z & n_x + in_y \\ n_x - in_y & 1 - n_z \end{bmatrix} \cdot \begin{bmatrix} \Psi_R \\ \Psi_R^c \end{bmatrix} \quad (2.7b)$$

One might expect that matrices from the previous two equations can be combined to produce the identity, but this is a false assumption. Instead, a pair of similar

eigenvalue equations emerge, depending on the order of the matrices

$$\begin{bmatrix} \mathcal{Z}_3 & \mathcal{Z}_1 + i\mathcal{Z}_2 \\ \mathcal{Z}_1 - i\mathcal{Z}_2 & -\mathcal{Z}_3 \end{bmatrix} \cdot \begin{bmatrix} \Psi_R \\ \Psi_R^c \end{bmatrix} = \eta \begin{bmatrix} \Psi_R \\ \Psi_R^c \end{bmatrix} \quad (2.8a)$$

$$\begin{bmatrix} n_z & n_x + in_y \\ n_x - in_y & -n_z \end{bmatrix} \cdot \begin{bmatrix} \psi_R \\ \psi_R^c \end{bmatrix} = \begin{bmatrix} \psi_R \\ \psi_R^c \end{bmatrix} \quad (2.8b)$$

Similar equations can also be derived for the left-handed states.

Unitary Matrices in the Electromagnetic Basis: Now suppose $\mathcal{H}_r$ is diagonalized using electromagnetic state vectors. Keeping particle and antiparticle pairs together once again, a new set of 2×2 special unitary matrices U_1 and U_2 are:

$$U_1 = \frac{1}{\sqrt{\Psi_R^\dagger \Psi_R}} \begin{bmatrix} \eta + \mathcal{Z}_3 & -\mathcal{Z}_1^* + i\mathcal{Z}_2^* \\ \mathcal{Z}_1 + i\mathcal{Z}_2 & \eta + \mathcal{Z}_3^* \end{bmatrix} \quad \Psi_R^\dagger \Psi_R = \Psi_L^{\dagger c} \Psi_L^c \quad (2.9a)$$

$$U_2 = \frac{1}{\sqrt{\Psi_L^\dagger \Psi_L}} \begin{bmatrix} \eta - \mathcal{Z}_3^* & \mathcal{Z}_1 - i\mathcal{Z}_2 \\ -\mathcal{Z}_1^* - i\mathcal{Z}_2^* & \eta - \mathcal{Z}_3 \end{bmatrix} \quad \Psi_L^\dagger \Psi_L = \Psi_R^{\dagger c} \Psi_R^c \quad (2.9b)$$

Normalizations are naturally more difficult and may be written:

$$\begin{aligned} \Psi_R^\dagger \Psi_R &= \eta^2 + \eta(\mathcal{Z}_3 + \mathcal{Z}_3^*) + \vec{\mathcal{Z}} \cdot \vec{\mathcal{Z}}^* + i\mathcal{Z}_1^* \mathcal{Z}_2 - i\mathcal{Z}_1 \mathcal{Z}_2^* \\ &= 2[E^2 + \eta E_z - (\mathbf{E} \times \mathbf{B})_z] \end{aligned} \quad (2.10a)$$

$$\begin{aligned} \Psi_L^\dagger \Psi_L &= \eta^2 - \eta(\mathcal{Z}_3 + \mathcal{Z}_3^*) + \vec{\mathcal{Z}} \cdot \vec{\mathcal{Z}}^* - i\mathcal{Z}_1^* \mathcal{Z}_2 + i\mathcal{Z}_1 \mathcal{Z}_2^* \\ &= 2[E^2 - \eta E_z + (\mathbf{E} \times \mathbf{B})_z] \end{aligned} \quad (2.10b)$$

Results similar to equation (1.20) are:

$$U_1^\dagger \mathcal{H}_r U_1 = \begin{bmatrix} \varepsilon_r & 0 \\ 0 & -\varepsilon_r \end{bmatrix} \quad U_2^\dagger \mathcal{H}_r U_2 = \begin{bmatrix} -\varepsilon_r & 0 \\ 0 & \varepsilon_r \end{bmatrix} \quad (2.11)$$

Characteristic equations associated with U_1 and U_2 may be written

$$U_1 : \quad \lambda^2 - 2\lambda \left[\frac{\eta + E_z}{\sqrt{2E^2(1 - n_z) + 4\eta E_z}} \right] + 1 = 0 \quad (2.12a)$$

$$U_2 : \quad \lambda^2 - 2\lambda \left[\frac{\eta - E_z}{\sqrt{2E^2(1 + n_z) - 4\eta E_z}} \right] + 1 = 0 \quad (2.12b)$$

To solve for λ it is useful to assume that particle velocity in both equations moves only in the z-direction $\vec{\beta} = (0, 0, \beta)$. This simplifies the characteristic equations which becomes identical to their neutrino counterparts in Table 2. This means that Table 2 applies equally to the neutrino problem and the electromagnetic problem. It should come as no surprise that the eigenstates of the electromagnetic problem are those of equation (1.21).

Noether Currents from Electromagnetic Eigenstates: Under the assumption that electromagnetic states are a viable replacement for the neutrino states, a question arises as to whether they can be used to generate Noether currents. To answer this question, suppose (with no loss of generality) that electromagnetic states are simplified for particle motion along the z-direction. In this scenario, for equivalent normalizations one finds

$$\lim_{\beta_x, \beta_y \rightarrow 0} \Psi_R = \psi_R \qquad \lim_{\beta_x, \beta_y \rightarrow 0} \Psi_L = \psi_L \qquad (2.13a)$$

$$\lim_{\beta_x, \beta_y \rightarrow 0} \Psi_R^c = \psi_R^c \qquad \lim_{\beta_x, \beta_y \rightarrow 0} \Psi_L^c = \psi_L^c \qquad (2.13b)$$

In other words, the two competing sets of eigenstates are really just the same thing, and neutrino Noether currents of (1.17) remain unchanged. A second possibility requires use of the inverse transformation in equation (2.6) to write the neutrino currents in terms of electromagnetic eigenstates. For ψ_R only, write

$$\begin{aligned} J_R^\nu &= \psi_R^\dagger \sigma^\nu \psi_R = \frac{1}{4\eta^2} \left[(1 + n_z)^2 \Psi_R^\dagger \sigma^\nu \Psi_R + (n_x^2 + n_y^2) \Psi_R^\dagger \sigma^\nu \Psi_R^c \right] \\ &+ \frac{1}{4\eta^2} \left[(1 + n_z)(n_x - in_y) \Psi_R^\dagger \sigma^\nu \Psi_R + (1 + n_z)(n_x + in_y) \Psi_R^\dagger \sigma^\nu \Psi_R^c \right] \end{aligned} \quad (2.14)$$

3 Hamiltonians for Weyl Neutrinos

For the retarded problem only, a Hamiltonian associated with (1.5a) follows by multiplying on the left with γ^0 . The 4×4 operator equation in the Dirac representation can be written

$$\mathcal{H}_r \psi_o = \varepsilon_r \psi_o \quad \text{where} \quad \mathcal{H}_r = \varepsilon_r \begin{bmatrix} 0 & \boldsymbol{\sigma} \cdot \hat{\mathbf{n}} \\ \boldsymbol{\sigma} \cdot \hat{\mathbf{n}} & 0 \end{bmatrix} \quad (3.1)$$

To build helicity eigenstates for this operator it is only necessary to double the size of the two component states of equation (1.12) with two well-placed minus signs in the left-handed states.

$$\phi_R = \frac{1}{\sqrt{2}} \begin{bmatrix} \psi_R \\ \psi_R \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \qquad \phi_L^c = \frac{1}{\sqrt{2}} \begin{bmatrix} \psi_L^c \\ -\psi_L^c \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad (3.2a)$$

$$\phi_L = \frac{1}{\sqrt{2}} \begin{bmatrix} \psi_L \\ -\psi_L \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \qquad \phi_R^c = \frac{1}{\sqrt{2}} \begin{bmatrix} \psi_R^c \\ \psi_R^c \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (3.2b)$$

As usual, the right-handed states have positive energy eigenvalues while the left-handed states have negative energy eigenvalues. All the labels remain unchanged from the two-component theory and may be verified with the charge conjugation operator $\mathbb{C}$.

Electromagnetic Hamiltonian: In the spherical basis, the Hamiltonian of (3.1) can be replaced by an equivalent electromagnetic counterpart. To derive it, apply the operator $\eta \cdot \gamma^\nu \beta_\nu$ to (1.5a):

$$\eta \cdot \gamma^\nu \beta_\nu \frac{\gamma^\nu R_\nu}{\rho} \psi_o = [\eta - \Delta_v] \psi_o = 0 \quad (3.3)$$

The new eigenvalue equation features a Hamiltonian operator Δ_v composed of charged particle velocity fields ($\mathbf{E}, \mathbf{B}$), and defined by

$$\Delta_v \psi_o = \eta \psi_o \quad \text{where} \quad \Delta_v \equiv \begin{bmatrix} i\boldsymbol{\sigma} \cdot \mathbf{B} & \boldsymbol{\sigma} \cdot \mathbf{E} \\ \boldsymbol{\sigma} \cdot \mathbf{E} & i\boldsymbol{\sigma} \cdot \mathbf{B} \end{bmatrix}_v \quad (3.4)$$

Eigenstates of Δ_v are still the neutrino eigenstates of (3.2), and this fundamentally associates Weyl neutrinos with particle Coulomb fields. Additionally, Δ_v is proportional to the square root of the classical electromagnetic Lagrangian. It is a non-Hermitian operator with its own spectrum of eigenstates and characteristic equation

$$(\mathbf{B}^2 + \lambda^2)^2 + \mathbf{E}^4 - 2\mathbf{E}^2 \lambda^2 - 2\mathbf{E}^2 \mathbf{B}^2 + 4(\mathbf{E} \cdot \mathbf{B})^2 = 0 \quad (3.5)$$

But $\mathbf{E} \cdot \mathbf{B} = 0$ for particle Coulomb fields, and this renders the set of real eigenvalues $\lambda = \pm\eta$. Degenerate eigenstates for Δ_v associated with $+\eta$ are:³

$$\Phi_a = \frac{1}{\sqrt{2}} \begin{bmatrix} \eta + iB_z \\ iB_x - B_y \\ E_z \\ E_x + iE_y \end{bmatrix} \quad \Phi_b = \frac{1}{\sqrt{2}} \begin{bmatrix} iB_x + B_y \\ \eta - iB_z \\ E_x - iE_y \\ -E_z \end{bmatrix} \quad (3.6a)$$

$$\Phi_c = \frac{1}{\sqrt{2}} \begin{bmatrix} E_z \\ E_x + iE_y \\ \eta + iB_z \\ iB_x - B_y \end{bmatrix} \quad \Phi_d = \frac{1}{\sqrt{2}} \begin{bmatrix} E_x - iE_y \\ -E_z \\ iB_x + B_y \\ \eta - iB_z \end{bmatrix} \quad (3.6b)$$

To determine Helicity eigenstates, add or subtract two well chosen state vectors in (3.6). Similar to the neutrino states of (3.2), one may write

$$\Phi_R = \frac{1}{\sqrt{2}} \begin{bmatrix} \Psi_R \\ \Psi_R \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad \Phi_L^c = \frac{1}{\sqrt{2}} \begin{bmatrix} \Psi_L^c \\ -\Psi_L^c \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad (3.7a)$$

$$\Phi_L = \frac{1}{\sqrt{2}} \begin{bmatrix} \Psi_L \\ -\Psi_L \end{bmatrix} \cdot e^{-ik(c\tau - \rho)} \quad \Phi_R^c = \frac{1}{\sqrt{2}} \begin{bmatrix} \Psi_R^c \\ \Psi_R^c \end{bmatrix} \cdot e^{+ik(c\tau - \rho)} \quad (3.7b)$$

³To derive eigenstates with eigenvalue $-\eta$, simply change the sign $\eta \rightarrow -\eta$ in each eigenstate of equation (3.6). Unfortunately, negative eigenvalue states are not legitimate and unphysical. Essentially, if the sign of η is changed in the eigenstate, the signs of all the fields must be changed with it since $\eta = \mathbf{E} \cdot \hat{\mathbf{n}}$. One may also demonstrate that $-\eta$ field eigenstates are not eigenstates of the neutrino operator in equation (3.1).

Classical Conservation Laws: An important use of Δ_v is for the derivation of conservation laws associated with classical electromagnetism:

$$\mathcal{T} \equiv \frac{1}{8\pi} \Delta_v \Delta_v^* = \begin{bmatrix} H & \boldsymbol{\sigma} \cdot \mathbf{S} \\ \boldsymbol{\sigma} \cdot \mathbf{S} & H \end{bmatrix} \quad (3.8)$$

where H is the Hamiltonian for the electromagnetic field and $\mathbf{S}$ is the Poynting vector. Now apply the derivative $\gamma^\nu \partial_\nu \mathcal{T}$ to generate the following two conservation equations

$$\frac{\partial H}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{S} + i\boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \times \mathbf{S} = 0 \quad (3.9a)$$

$$\boldsymbol{\sigma} \cdot \left[\frac{\partial \mathbf{S}}{\partial t} + \boldsymbol{\nabla} H \right] = -\boldsymbol{\sigma} \cdot \mathbf{J} \quad (3.9b)$$

Two Component Hamiltonians: Similar to equation (3.3), the two-component neutrino operator equations of (1.10) can also be transformed into Coulomb field operators by removing the energy ε_r and writing

$$[\eta \cdot \bar{\sigma}^\nu \beta_\nu] \sigma^\nu n_\nu \cdot \psi_R = (\eta - \boldsymbol{\sigma} \cdot \mathbf{Z}) \psi_R = 0 \quad (3.10)$$

$$[\eta \cdot \sigma^\nu \beta_\nu] \bar{\sigma}^\nu n_\nu \cdot \psi_L = (\eta + \boldsymbol{\sigma} \cdot \mathbf{Z}^*) \psi_L = 0 \quad (3.11)$$

A pair of 2×2 electromagnetic left/right Hamiltonian eigenvalue problems are:

$$\mathcal{H}_R \psi_R \equiv \boldsymbol{\sigma} \cdot \mathbf{Z} \psi_R = \eta \psi_R \quad \mathcal{H}_L \psi_L \equiv -\boldsymbol{\sigma} \cdot \mathbf{Z}^* \psi_L = \eta \psi_L \quad (3.12)$$

As before, neutrino eigenstates in both equations can be exchanged for their electromagnetic counterparts. Additionally, conservation laws may be calculated by writing:

$$\frac{1}{8\pi} \mathcal{H}_R \cdot \mathcal{H}_L = \frac{1}{8\pi} \boldsymbol{\sigma} \cdot \mathbf{Z} \cdot \boldsymbol{\sigma} \cdot \mathbf{Z}^* = H + \boldsymbol{\sigma} \cdot \mathbf{S} \quad (3.13)$$

Then apply the para-gradient operator

$$(\partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})(H + \boldsymbol{\sigma} \cdot \mathbf{S}) = -\boldsymbol{\sigma} \cdot \mathbf{J} \quad (3.14)$$

4 Electromagnetic Noether Currents

As Weyl neutrinos carry energy and momentum determined by their Noether currents, it is sensible to suggest that a theory of electron Coulomb fields might also be constructed as carriers of momentum and energy. But as previously shown, construction of Noether currents with electromagnetic eigenstates simply re-produce neutrino currents. This result was not surprising either since the use of electromagnetic states is simply a change of basis for addressing the Weyl neutrino problem. But a method

for developing a theory of a Coulomb radiation currents is still possible, and may be constructed by making full use of the harmony which exists between the two theories.

This harmony dictates that all properties of Weyl neutrinos have an associated electromagnetic counterpart in the spherical basis. To derive an electromagnetic current J_{EM}^ν from the neutrino current $J_{wyl}^\nu = \varepsilon_r(1, \hat{\mathbf{n}})$, one must therefore construct a 4×4 tensor $M^\mu{}_\nu$ representing the four velocity of an electron, capable of deriving an appropriate J_{EM}^ν through the tensor relation $J_{EM}^\mu = M^\mu{}_\nu \cdot J_{wyl}^\nu$. A step towards accomplishing this is to write the relation which derives the electromagnetic paravector of (2.1) in terms of the null vector $(1, \hat{\mathbf{n}})$:

$$\begin{bmatrix} \eta \\ \mathbf{E}_x + i\mathbf{B}_x \\ \mathbf{E}_y + i\mathbf{B}_y \\ \mathbf{E}_z + i\mathbf{B}_z \end{bmatrix} = \frac{\eta}{(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta})} \begin{bmatrix} 1 & -\beta_x & -\beta_y & -\beta_z \\ -\beta_x & 1 & -i\beta_z & i\beta_y \\ -\beta_y & i\beta_z & 1 & -i\beta_x \\ -\beta_z & -i\beta_y & i\beta_x & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ n_x \\ n_y \\ n_z \end{bmatrix} \quad (4.1)$$

This relation brings the theory of Weyl neutrinos full circle with its electromagnetic counterpart. If the neutrino energy is included in (4.1), it can be written more succinctly as

$$\varepsilon \mathcal{Z}^\mu = \eta M^\mu{}_\nu J_{wyl}^\nu \quad (4.2)$$

By dividing out the dilatation and including a similar complex conjugated equation, this will generate two polarization four-vectors for the fields applicable to both electrons and positrons.

$$\begin{cases} z_R^\nu = (1, \hat{\mathbf{z}}) \\ z_L^\nu = (1, \hat{\mathbf{z}}^*) \end{cases} \quad \text{where} \quad \hat{\mathbf{z}} \equiv \frac{\gamma}{\mathcal{D}_-} [\hat{\mathbf{n}} - \boldsymbol{\beta} + i\boldsymbol{\beta} \times \hat{\mathbf{n}}] \quad (4.3)$$

For each polarization, electric and magnetic components are out of phase by $\pm\pi/2$ similar to the problem of circularly polarized light. Now suppose the neutrino energy is re-written $\varepsilon = cp = \hbar c/r_e$ where r_e is (possibly) the classical electron radius. Two forms of particle momentum null paravectors are then

$$p_R^\nu \equiv pz_R^\nu = [p, \mathbf{p}] \quad p_L^\nu \equiv pz_L^\nu = [p, \mathbf{p}^*] \quad (4.4)$$

An associated complex wave vector $\mathbf{k} = \hat{\mathbf{z}}/r_e$ implies a particle wavelength $\lambda = 2\pi r_e$. In large numbers, radiated particles are expected to re-produce the classical results

$$\boldsymbol{\pi}_E = \dot{N}_{em} \cdot \frac{1}{2} [\mathbf{p} + \mathbf{p}^*] \quad \boldsymbol{\pi}_B = \dot{N}_{em} \cdot \frac{1}{2i} [\mathbf{p} - \mathbf{p}^*] \quad (4.5)$$

where $\dot{N}_{em}$ is the particle number flux, or number of particle emissions per unit proper time per unit area. Using the fundamental Dirac frequency ω_D , this number is Lorentz invariant and proportional to the dilatation

$$\dot{N}_{em} = \frac{\omega_D}{2\pi\rho^2} = \frac{dN}{d\tau dA} \quad (4.6)$$

A momentum flux version of the classical field strength tensor (less particle accelerations) is

$$F_{\pi}^{\mu\nu} = \frac{\dot{N}_{em}}{2} [P^{\mu\nu} + P^{*\mu\nu}] \quad (4.7)$$